Modeling and Forecasting Monthly Atmospheric CO_2 at Mauna Loa, 1958–2026

Final Project Report

Pascal Antwi · STA6938 Time Series · Instructor: Y. Jia · December 6, 2026

Abstract. Atmospheric carbon dioxide (CO_2) is the principal driver of anthropogenic climate change, and its rate of increase is central to climate policy. This study models and forecasts the 822 monthly mean concentrations recorded at Mauna Loa Observatory from March 1958 to August 2026. Time plots, classical decomposition, sample autocorrelations with 95% confidence bands and augmented Dickey–Fuller tests reveal an accelerating stochastic trend and a stable additive 12-month cycle, so both regular and seasonal differencing are required. Four model families (ARMA, ARIMA, seasonal ARIMA and regression with autocorrelated errors) were compared using ACF/PACF identification, the Schwarz Bayesian criterion, residual diagnostics and rolling-origin forecast evaluation. The selected ARIMA$(0,1,1)\times(0,1,1)_{12}$ model has highly significant parameters ($\hat{\theta}_1 = 0.389$, $\hat{\Theta}_1 = 0.870$), is invertible, and leaves white-noise residuals with constant variance; it is the most accurate family out of sample (RMSE 0.55 ppm, 96% interval coverage over 276 forecasts). Growth has tripled from 0.86 to 2.54 ppm per year since the 1960s, and monthly CO_2 is forecast to exceed 440 ppm for the first time in April 2030 (95% prediction interval 437.8–443.9 ppm).

Keywords: seasonal ARIMA; ARMA; regression with autocorrelated errors; Holt–Winters; Mauna Loa.

1. Background and Motivation

Carbon dioxide (CO_2) is the most important long-lived greenhouse gas, and its rising atmospheric concentration is the principal driver of anthropogenic climate change. The Mauna Loa record, initiated by Charles D. Keeling in 1958, is the longest continuous direct measurement of atmospheric CO_2 [1], and international climate targets are framed around the rate at which concentrations rise and the dates at which particular thresholds are crossed. A rigorous statistical characterization and forecast of this series is therefore of direct scientific and policy relevance. The series is also well suited to time series methodology, as it combines a stochastic trend, a pronounced seasonal cycle and short-term serial dependence within a long and complete record.

Although the upward trend in atmospheric CO_2 is well established, its current rate of increase, the evidence for acceleration, and the levels expected over the coming years require quantification with appropriate measures of statistical uncertainty. This study therefore pursues four objectives: (i) to characterize the trend and seasonality of monthly CO_2; (ii) to build and validate an adequate time series model, comparing ARMA, ARIMA, seasonal ARIMA and regression with autocorrelated errors; (iii) to assess whether growth is accelerating; and (iv) to forecast monthly CO_2 to 2030 with prediction intervals, including the date at which it first exceeds 440 ppm.

2. Description of the Series

The data are the monthly mean CO_2 mole fractions, in parts per million (ppm), published by the NOAA Global Monitoring Laboratory [2] and freely available at https://gml.noaa.gov/ccgg/trends/data.html. The series X_t contains $n = 822$ consecutive months from March 1958 to August

2026, with no missing values (Figure 1 and Table 1). Concentration rose from 315.71 ppm in March 1958 to 427.55 ppm in August 2026, an increase of 111.8 ppm (35%). Annual growth averaged 1.68 ppm but tripled, from 0.86 ppm/yr in the 1960s to 2.40 ppm/yr in the 2010s and 2.54 ppm/yr in the 2020s (Figure 2 and Table 2).

A classical additive decomposition with a 2×12 moving-average filter [3] (Figure 3) attributes 96.6% of the variance of X_t to the trend. The seasonal component peaks in May (+3.04 ppm) and reaches its minimum in October (-3.25 ppm), an amplitude of 6.29 ppm (Figure 4 and Table 3).

3. Stationarity of the Series

Time plot. The first step was the time plot (Figure 1), which shows two clear features: a persistent upward trend, and a regular 12-month seasonal cycle with nearly the same shape every year. The mean of X_t therefore depends on t, and the series is non-stationary.

Transformation. A log or Box–Cox transformation is warranted when variability grows with the level [3]. The within-year range of the detrended series (Figure 5) stays between 5.3 and 7.6 ppm (6.0 ppm in the 1960s, 7.0 ppm in the 2010s) while the level rose 35%, so the additive form is adequate and no transformation is used.

Sample ACF. The sample ACF agrees (Figure 6): it decays only very slowly, from $\hat{\rho}(1) = 0.996$ to 0.865 at lag 36, and stays far outside the 95% confidence band (± 0.068) at every lag, the classic signature of a unit root. After a first difference, ∇X_t, the ACF at the seasonal lags 12, 24 and 36 remains 0.925, 0.910 and 0.894. This failure to decay shows that the seasonal pattern is also non-stationary, so a seasonal difference $\nabla_{12} = 1 - B^{12}$ is required. The seasonally differenced series $\nabla_{12} X_t$ is the 12-month change in concentration. Its ACF still decays slowly (0.856, 0.792 and 0.745 at lags 1–3), and its level rises steadily over time as the growth rate accelerates, so it does not have a constant mean.

Augmented Dickey–Fuller tests. Because seasonal swings mimic mean reversion, the ADF test [4] was applied to the seasonally adjusted series (Table 4). With a single mean, τ ranges from 3.57 to 4.85, far above the 5% critical value of -2.86; with a trend, τ ranges from -0.88 to -0.62, against a critical value of -3.41. A unit root is never rejected. For $\nabla_{12} X_t$ the test rejects (τ from -7.91 to -5.37), but an AR(1)-plus-drift model for $\nabla_{12} X_t$ estimates $\hat{\phi} = 0.9993$ (SE 0.0010), practically a unit root. For the doubly differenced series $\nabla \nabla_{12} X_t$, the test rejects decisively at every lag (τ from -37.89 to -22.05). I therefore model $W_t = (1 - B)(1 - B^{12}) X_t$, which has 809 observations, as a stationary series (Figure 7).

Estimation versus differencing. As an alternative to differencing, I estimated the deterministic components by regression on a quadratic trend and 11 monthly dummy variables (Table 5). The fit is excellent ($R^2 = 0.9994$, residual SD 0.82 ppm), and the positive quadratic coefficient confirms that the trend is accelerating. The residuals, however, are far from white noise (Figure 8): their lag-1 autocorrelation is 0.915, $Q(12) = 5991$, and the Durbin–Watson statistic is $D = 0.17$, far below the value of 2 expected for independent errors. Differencing is therefore more appropriate; regression with ARMA errors is evaluated in Section 4.

4. Model Selection and Model-Building Procedure

The model was built with the Box–Jenkins procedure [5] (plot, transform, identify, estimate, diagnose, forecast), with $d = D = 1$ and $T = 12$ fixed in Section 3.

White-noise test. The Ljung–Box test [6] for W_t gives $Q(6) = 69.99$, $Q(12) = 283.32$, $Q(18) = 295.43$ and $Q(24) = 301.19$ (all $p < 0.001$), so W_t is not white noise and must be modeled (Table 6).

Identification. The orders were identified from the overall pattern of the sample ACF and PACF of W_t (Figure 9), using the 95% confidence band $\pm 1.96/\sqrt{809} = \pm 0.069$ as a guide to individual significance and the Ljung–Box test above for joint significance. The ACF is dominated by two large spikes, at lag 1 (-0.280) and the seasonal lag 12 (-0.471), with a satellite at lag 11 (0.185), and then drops to near zero; the borderline values at lags 14 (0.088) and 23 (-0.069) are isolated and consistent with the two or three exceedances expected by chance among 48 lags, so the ACF cuts off. The PACF instead decays gradually, both over lags 1–3 (-0.280, -0.146, -0.135) and across the seasonal lags 12, 24 and 36 (-0.378, -0.260, -0.163), so it tails off. An ACF that cuts off with a PACF that tails off is the signature of moving-average behavior [3], here at both lag 1 and the seasonal lag 12. This points to the multiplicative seasonal ARIMA model [5]:

$$(1 - B)(1 - B^{12})X_t = (1 - \theta_1 B)(1 - \Theta_1 B^{12})Z_t, \qquad Z_t \sim \mathrm{WN}(0, \sigma_Z^2), \tag{1}$$

that is, ARIMA$(0,1,1) \times (0,1,1)_{12}$. Its right-hand side is an MA(13) with only two free parameters, which explains the small autocorrelations at lags 11 and 13.

Estimation and selection. Nine seasonal ARIMA candidates (Table 7) were estimated by maximum likelihood (ML) and compared with AIC and SBC [7]. ARIMA$(0,1,1) \times (0,1,1)_{12}$ attains the lowest SBC (465.39); ARIMA$(1,1,1) \times (0,1,1)_{12}$ has a lower AIC (447.58 versus 451.30) but a higher SBC (466.37). Pure AR alternatives fit worse: ARIMA$(1,1,0) \times (0,1,1)_{12}$ and ARIMA$(2,1,0) \times (0,1,1)_{12}$ have SBC 486.92 and 479.42. Adding seasonal AR terms raises the SBC by at least 6.7, and ARIMA$(0,1,1) \times (1,1,0)_{12}$ is clearly inadequate (SBC 683.43; residuals fail the Ljung–Box test). Because SBC penalizes extra parameters more heavily in large samples, the two-parameter model was selected. Its implied autocorrelations ($\rho(1) = -0.338$, $\rho(12) = -0.495$, $\rho(11) = 0.167$) match the sample values.

Comparison of the four model families. Four model families were fitted, each with its order chosen by SBC (Table 8 and Figure 10). (a) *ARMA*: the best non-seasonal model for W_t, ARMA(3,2), leaves a residual autocorrelation of -0.409 at lag 12 (Ljung–Box $p < 0.001$), because a non-seasonal ARMA cannot represent the seasonal MA structure. (b) *ARIMA*: the best ARIMA$(p,1,q)$ with drift, ARIMA(4,1,4), has four AR roots of modulus 1.000–1.001 with periods of 6 and 12 months. It imitates a seasonal difference with unit roots, and its residuals remain correlated at lag 12 (0.180, $p < 0.001$). (c) *Regression with autocorrelated errors*: the quadratic trend and monthly dummies with ARMA(2,1) errors give an AR root of 1.02, almost a unit root, and residual correlation of 0.138 at lag 12 ($Q(12)$ $p < 0.001$; Table 9). The standard remedy is then to difference both sides at lags 1 and 12, which removes the regressors and leads back to a seasonal ARIMA model. (d) *Seasonal ARIMA*: only ARIMA$(0,1,1) \times (0,1,1)_{12}$ leaves white-noise residuals (lag-12 autocorrelation -0.005, $Q(12)$ $p = 0.464$). The seasonal ARIMA family is therefore required.

5. Hypothesis Tests and Confidence Intervals

The ML estimates (Table 10) are $\hat{\theta}_1 = 0.3885$ (SE 0.0367) and $\hat{\Theta}_1 = 0.8700$ (SE 0.0170), with $\hat{\sigma}_Z^2 = 0.1012$, so $\hat{\sigma}_Z = 0.318$ ppm. The log-likelihood is -222.65. For $H_0 : \theta_1 = 0$, $z = 10.59$ ($p < 0.001$), with 95% confidence interval $[0.317, 0.460]$. For $H_0 : \Theta_1 = 0$, $z = 51.28$ ($p < 0.001$), with 95% interval $[0.837, 0.903]$. Both terms are therefore essential. Conditional least squares gives similar estimates ($\hat{\theta}_1 = 0.417$, $\hat{\Theta}_1 = 0.812$), so the conclusions do not depend on the estimation

method.

The root of $1 - \hat{\theta}_1 B$ is $1/\hat{\theta}_1 = 2.57$, and the roots of $1 - \hat{\Theta}_1 B^{12}$ have modulus $\hat{\Theta}_1^{-1/12} = 1.012$. Both lie outside the unit circle, so W_t is *invertible*, and with no AR part it is also causal; the model can therefore be used for forecasting. The interval for Θ_1 lies entirely below 1; $\Theta_1 = 1$ would indicate a deterministic seasonal pattern, so seasonal differencing is justified. The large $\hat{\Theta}_1$ means the seasonal pattern changes only slowly from year to year, and $\hat{\theta}_1$ means about 39% of each month's shock is reversed the following month.

6. Model Adequacy Checking and Remedial Methods

Independence. The residuals (Figure 11) pass the Ljung–Box test at all four lags examined (Table 11 and Figure 12): $Q(6) = 9.05$ ($p = 0.060$), $Q(12) = 9.74$ ($p = 0.464$), $Q(18) = 14.43$ ($p = 0.567$) and $Q(24) = 20.83$ ($p = 0.532$), with degrees of freedom reduced by the two estimated parameters. The residual ACF shows no systematic pattern: only lag 3 (-0.081) lies just outside the 95% band of ± 0.069, fewer than the 1.8 exceedances expected by chance among 36 lags.

Constant variance. The Ljung–Box test on the squared residuals gives $Q(12) = 12.40$ ($p = 0.414$) and $Q(24) = 35.36$ ($p = 0.063$). The residual variance is similar in both halves of the sample (0.091, 0.107), so it is constant.

Normality. The residuals are nearly symmetric (skewness 0.11, excess kurtosis 0.35), but three tests reject normality at 5%: Shapiro–Wilk $W = 0.9963$ ($p = 0.048$), Anderson–Darling ($p = 0.038$) and Jarque–Bera ($p = 0.006$). Five months have standardized residuals beyond ± 3 (Table 12): March 1959 (-3.46), March 1977 (3.31), September 1986 (3.11), April 2016 (3.97) and March 2022 (-3.53). April 2016 coincides with the 2015–16 El Niño.

Remedial methods.

(i) *Lag-3 autocorrelation.* To remove this correlation, I added an AR(1) term. ARIMA$(1,1,1) \times (0,1,1)_{12}$ has a significant $\hat{\phi}_1 = 0.215$ (SE 0.084, $p = 0.010$) and reduces the lag-3 residual autocorrelation to -0.042. However, it raises the SBC and, in the hold-out of Section 7, forecast less accurately (RMSE 1.168 versus 1.110) with only 83% interval coverage against 97%. The parsimonious model was retained.

(ii) *Non-normality.* The five outlying months were treated as missing and the model was re-estimated by exact likelihood (Kalman filter). The estimates were almost unchanged (Table 13; $\hat{\theta}_1 = 0.3838$, $\hat{\Theta}_1 = 0.8689$), and normality was no longer rejected (Shapiro–Wilk $p = 0.211$, Anderson–Darling $p = 0.101$, Jarque–Bera $p = 0.094$). The residuals still passed the Ljung–Box tests ($p \geq 0.092$). The non-normality is thus caused by a few genuine but unusual months; with $n = 822$ the ML estimates are approximately normal, so the tests and intervals remain valid.

Because the residual variance is stable, no variance-stabilizing transformation is needed, and the fitted ARIMA$(0,1,1) \times (0,1,1)_{12}$ model is adequate.

7. Forecasting

Rolling-origin comparison. Each method was re-estimated at 23 forecast origins (every August from 2000 to 2022) and forecast the next 12 months, giving 276 out-of-sample forecasts per method (Figure 13 and Table 14). The seasonal ARIMA model achieved RMSE 0.552 ppm, with 95% interval coverage of 96.0%. It was essentially tied with Holt–Winters exponential smoothing [8]

(0.543 ppm, 96.0%), and more accurate than regression with ARMA errors (0.580 ppm, 93.8%), the ARMA model for W_t (0.677 ppm) and non-seasonal ARIMA (0.697 ppm). Holt–Winters estimates $\hat{\alpha} = 0.534$, $\hat{\beta} = 0.014$, $\hat{\gamma} = 0.299$ (Table 15).

Recent hold-out. Fitted through August 2023 and forecasting September 2023 to August 2026 (Figure 14 and Table 16), the seasonal ARIMA model had RMSE 1.110 ppm, MAPE 0.24% and 97.2% interval coverage. Holt–Winters had RMSE 0.981 ppm, the trend-and-dummies regression 1.491 ppm (only 52.8% coverage, as it ignores the residual autocorrelation) and the seasonal naive method 6.448 ppm. All methods under-predicted 2024, when the annual mean rose by a record 3.52 ppm during the 2023–24 El Niño.

Forecasts to 2030. The model estimated on all 822 months (Figure 15 and Table 17) gives a forecast for September 2026 of 426.16 ppm, with 95% prediction interval [425.53, 426.78]. The forecast first exceeds 440 ppm in April 2030 (440.84 ppm [437.78, 443.89]) and peaks at 441.59 ppm [438.49, 444.69] in May 2030. The forecast for December 2030 is 439.03 ppm [435.59, 442.47]. Forecast annual means rise from 431.44 ppm (2027) to 438.67 ppm (2030), about 2.4 ppm per year, and the 95% interval widens from 1.25 ppm to 6.88 ppm, reflecting the accumulated uncertainty of a doubly integrated process.

8. Conclusions

Monthly CO_2 at Mauna Loa is a non-stationary series with an accelerating stochastic trend and a stable additive seasonal cycle. Of the four model families compared (ARMA, ARIMA, seasonal ARIMA, and regression with autocorrelated errors), only the seasonal ARIMA leaves white-noise residuals and gives the most accurate rolling forecasts, and the ACF/PACF and information criteria point to the parsimonious ARIMA$(0,1,1) \times (0,1,1)_{12}$. Growth has tripled since the 1960s, to about 2.5 ppm per year. The model projects that concentrations will cross 440 ppm in spring 2030. As a univariate model it cannot anticipate El Niño-driven surges such as those of 2016 and 2024; adding climate indices or emissions data is a natural extension.

Supplementary material. Click a figure or table number to open it online; all are in one supplementary PDF.

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